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Predicting Maximum Acceptable Efforts for Repetitive Tasks: An Equation Based on Duty Cycle

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Objective: The objective was to develop an equation, for repetitive tasks, that uses frequency and/or duty cycle (DC) to predict maximum acceptable efforts (MAE) relative to maximum voluntary efforts (MVE).

Background: Ergonomists must determine acceptable physical demands for a wide variety of tasks. Although a large database exists in the literature for maximum single-effort strength, far fewer repetitive tasks have psychophysical and/or physiological data available to guide the prediction of acceptable submaximal, repeated efforts.

Method: DC represents the total effort duration divided by the cycle time. MAEs were calculated by dividing average psychophysics-based acceptable loads by corresponding single-effort maximum strength using 69 values from studies of the upper extremities. The author developed an equation to characterize the relationship between MAE and DC.

Results: The resulting equation had DC taken to the exponent 0.24, and it predicted MAE very well ($r^2 = 0.87\%$, root mean square [RMS] difference = 7.2% of the maximum strength). At higher DC values, the equation also demonstrated good agreement with the published physiological data.

Conclusion: The limited psychophysical database in the literature makes it difficult for ergonomists and engineers to recommend acceptable efforts for the large variety of repetitive tasks they evaluate. However, the proposed equation now allows for a correction of the large strength database to estimate acceptable force and torque limits for repetitive occupational tasks.

Application: The proposed equation will have wide applications for ergonomic practitioners performing evaluations of repetitive tasks.

Keywords: physical work, loading, models and measures, fatigue

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INTRODUCTION

Ergonomists must evaluate a wide variety of workplace demands and compare them to worker capacities to establish tolerance limit values (TLVs). The demands are specific to the particular task, in terms of the postures, required grips and/or pinches, load distribution, force and/or torque requirements, and so on. We can often extract capacities from the literature for a particular population. Most often, the TLVs are set to be acceptable to 75% of females (25th percentile value), which also accommodates most males (Waters, Putz-Anderson, Garg, & Fine, 1993). In rare cases, workers perform a particular task once or twice a day, such that the recommended forces and torques can be very close to those recorded during single-effort, maximum voluntary efforts (MVEs) for the target population. MVEs result in maximum voluntary forces and/or torques that reflect the maximum strength for that particular condition. A large database exists in the literature to provide information regarding maximum strength for a wide variety of tasks (e.g., DiDomenico & Nussbaum, 2003; Greig & Wells, 2004; Mital & Kumar, 1998; Peebles & Norris, 2003). In addition, numerous ergonomics software packages incorporate existing strength data.

Although strength data can inform upper limits for single efforts, most occupational tasks require repetitive efforts over the course of a day. In such cases, the recommended demands should be lower than the maximum strength. For example, in auto assembly plants, the cycle times are usually close to 60 s, such that, even if the frequency of a particular effort is only one per cycle, the worker performs about 400 of these efforts each workday. It is obvious that a muscle, or joint, should not perform that many MVEs each day.

The power law in psychophysics was presented by Stevens (1957) to characterize the

exponential relationship between sensations and their physical stimuli. Sensation was shown to be proportional to the stimulus raised to some power, and these powers vary for different stimuli. Snook (1999) states that muscular effort and force follow the power law with an exponent of approximately 1.6. Snook (1968) performed one of the first published studies in ergonomics to take advantage of this relationship and use the psychophysical methodology to determine acceptable demands for repetitive tasks. He asked male respondents to determine maximum frequencies for lifting tasks with various set load weights and starting heights. Ayoub and Mital (1989) and Snook and Ciriello (1991) summarized many years of manual materials handling research conducted at Texas Tech University and Liberty Mutual Insurance, respectively. Those studies determined maximum acceptable forces and weights for many different lifting, lowering, pushing, pulling, and carrying tasks.

More recently, similar studies established guidelines for hand tool use (e.g., Kim & Fernandez, 1993), hand impacts (Potvin, Chiang, Mckean, & Stephens, 2000), and various other upper extremity demands (see the following review). One barrier to conducting these studies is that they take much longer to complete, compared to strength studies, especially when sufficient training is involved. As a result, there are far fewer published psychophysical studies related to acceptable work demands, and there are still many tasks that do not have data to guide the determination of acceptable forces and torques for repetitive efforts.

The publication of more future psychophysical studies will fill in the many gaps that currently exist in the ergonomics literature. However, in lieu of those data currently existing for many tasks, this study determines if there are any generalizable patterns that exist in the data from previously published studies. The purpose of this article was to perform a meta-analysis on existing psychophysical data to determine if I could develop an equation, using frequency, effort duration, and/or duty cycle as the only input or the inputs, to predict maximum acceptable efforts relative to MVE. Ergonomists and engineers could use such an equation to scale existing strength data to set

TLVs for specific occupational tasks that currently do not have psychophysical and/or physiological data to guide the determination of repetitive force and torque limits.

METHOD

Definitions

For the purposes of this article, maximum acceptable effort (MAE) refers to an average maximum acceptable force (MAF) or torque (MAT) that is normalized as a percentage of the average maximum force or torque (i.e., strength) from an MVE for a specific task. The psychophysical studies, used in ergonomics, control a number of variables and allow the respondent to set acceptable levels of force, torque, rest period, or frequency, usually based on an 8-hr workday. Each task typically requires one type of effort against resistance (e.g., hand grip, load lift, wrist flexion, etc.), and many publications provide maximum acceptable demands for more than one task. I treat each separately. Frequency is the number of discrete efforts per minute. Duty cycle (DC) is the percentage of time an individual is engaged in those efforts. For example, if a worker performs 10 thumb push efforts within each 65-s cycle and each effort is 0.55 s in duration, then the DC is $(10 \text{ efforts/cycle} \times 0.55 \text{ s/effort})/65 \text{ s per cycle}$, or $DC = 0.085$ (8.5% of each cycle).

Inclusion Criteria

A review of the literature provided a wide variety of different psychophysical studies related to ergonomics. For subsequent analyses, I developed strict and rigorous inclusion criteria for these studies. I placed a premium on respondents being provided with (a) sufficient training on each task and (b) an appropriate psychophysical adjustment period to determine the MAEs. Inclusion criteria are described in the following. The first three reflect the scope of the equation that was developed.

1. Respondents determined some maximum acceptable work demand on the upper extremities (e.g., torque, force, frequency etc) with other task characteristics being controlled.
2. Respondents were instructed to set these demands such that they would be acceptable for an 8-hr day.

3. The task did not include assistive devices (e.g., hand tools).
4. Each task was performed on multiple days to allow for sufficient training and to provide feedback on any delayed onset soreness that might result from selecting demands that were too high.
5. Respondents could determine acceptable demands for only one task on any particular test day. Ciriello, Snook, Webster, and Dempsey (2001) had respondents determine acceptable demands for six different tasks on one day, but concluded, "This constant rotation through tasks that should have separate psychophysical set points seems to have made the subjects gravitate toward a set point closer to the limiting movement" (p. 929). Thus, it was concluded that testing multiple conditions in one session might result in underestimated MAT or MAF values for some conditions.
6. The psychophysical adjustment period, on the final test day, must have been at least 40 min in duration. Muppasani and Fernandez (1996) performed a psychophysical study of a drilling task and found that using only a 20-min adjustment period resulted in overestimates of MAF by 39% to 48% when considering what would actually be acceptable for an 8-hr day. Wu and Chen (1991) studied the effects of adjustment periods of 20, 30, 40 and 50 min on maximum acceptable weights of lift (MAWL). They concluded that 20- and 30-min adjustment periods resulted in MAWL values that were overestimated by 9.5% and 12.4%, respectively, compared to those at 40 and 50 min (which did not differ).
7. The publication must have provided sufficient information to allow for the determination, or at least a good approximation, of effort duration to allow for the calculation of DC.

These criteria resulted in the inclusion of eight psychophysical studies of manual upper extremity tasks that are described and summarized in Table 1.

Upper Extremity Psychophysical Studies

The following describes a total of eight psychophysical studies presented in seven publications. Each study evaluated MATs and/or MAFs, and there were a total of 69 tasks and 117 respondents in these studies.

Snook, Vaillancourt, Ciriello, and Webster (1995) performed a study to determine MATs for concentric wrist flexion and extension efforts with various grips. The study was completed in two phases as they tested respondents either 2 or 5 days per week. For the 2 days per week study, female respondents ($n = 15$) performed (a) wrist flexion with power grip, (b) wrist flexion with pinch grip, and (c) wrist extension with power grip at $f = 2, 5, 10, 15$, and 20/min. There were five training days, starting at 2 hr per day and ramping up to 7 hr per day. They collected MATs during a 7-hr session on Days 6 thru to 20. They observed torque strength to remain consistent across sessions, so they calculated combined MVE means across the five frequencies and found these to be 7.24 Nm (wrist flexion with power grip), 6.15 Nm (wrist flexion with pinch grip), and 5.09 Nm (wrist extension with power grip; see their Table 2). For the 5 days per week study, female respondents ($n = 14$) performed wrist flexion with a power grip at $f = 15$ /min. There were four training days, starting at 4 hr per day and ramping up to 7 hr per day. They collected MATs during 7-hr sessions, from Days 5 to 23. There was no trend observed across days, for the ratio between MAT and MVE torque, so I used their grand means of 2.11 and 6.82 Nm, for MAT and MVE, respectively. They quantified average effort durations ($M = 0.81$ s) for only this one condition (see their Table 8). Since all conditions in Snook et al. (1995) required an angular displacement of 90 degrees and in lieu of other durations being presented, I assumed that the effort durations were also 0.81 s for each of the tasks in the 2 days per week study.

Snook, Vaillancourt, Ciriello, and Webster (1997) performed a study in two parts to determine MATs for ulnar deviation. Only Part B was considered, as it used a faster computer that gave respondents more time to complete each motion and allowed for higher, and more, frequencies to be tested. Female respondents ($n = 11$) performed concentric ulnar deviation efforts with a power grip at $f = 15, 20$ and 25/min. There were four training days, starting at 2 hr per day and ramping up to 6 hr per day. Maximum isometric strength was not observed to differ substantially across the three frequencies tested so I used the

TABLE 1: Summary of the Number of Respondents (total = 117), Task Frequencies, Durations, Duty Cycles (in s/min and as a percentage), and Tasks (total = 69) Within Each Study

Study		Freq (/min)	Duration (s)	Duty Cycle (s/min)	Duty Cycle (%)	Tasks	Respondents
Snook et al. (1995)	2 days/week	2	0.810	1.62	2.7	Wrist flexion: power grip;	15
		5	0.810	4.05	6.8	wrist flexion: pinch grip;	
		10	0.810	8.10	13.5	wrist extension: power	
		15	0.810	12.15	20.3	grip (n = Total of 15 tasks)	
		20	0.810	16.20	27.0		
	5 days/week	15	0.810	12.15	20.3	Wrist flexion: power grip (n = 1)	14
Snook et al. (1997)		15	1.120	16.80	28.0	Ulnar deviation: power grip	11
		20	1.050	21.00	35.0	(n = 3)	
		25	1.080	27.00	45.0		
Snook et al. (1999)		15	0.640	9.60	16.0	Wrist extension: pinch grip	20
		20	0.610	12.20	20.3	(n = 3)	
		25	0.550	13.75	22.9		
Ciriello et al. (2002)	Ulnar deviation	15	1.080	15.00	27.0	Ulnar deviation: power grip	10
		20	1.080	20.00	36.0	(n = 3); supination: 31-mm	
		25	1.080	25.00	45.0	screwdriver; supination:	
	Pronation and supination	15	1.215	15.00	30.4	40-mm screwdriver;	
		20	1.215	20.00	40.5	supination: 39-mm yoke;	
		25	1.215	25.00	50.6	pronation: 31-mm screwdriver (n = 12)	
Moore and Wells (2005)	Duty Cycle = 25%	3	5.000	15.00	25.0	Wrist extension: in-line	8
		5	3.000	15.00	25.0	powered screwdriver	
		10	1.500	15.00	25.0	(n = 11)	
	Duty Cycle = 50%	3	10.000	30.00	50.0		
		5	6.000	30.00	50.0		
		10	3.000	30.00	50.0		
		20	1.500	30.00	50.0		
	Duty Cycle = 83.3%	3	16.667	50.00	83.3		
		5	10.000	50.00	83.3		
		10	5.000	50.00	83.3		
		20	2.500	50.00	83.3		
Potvin et al. (2006)		2	0.160	0.32	0.5	Push: pulp pinch; push:	24
		7	0.160	1.12	1.9	finger tip (n = 6)	
		12	0.160	1.92	3.2		
Andrews et al. (2008)		1	0.685	0.69	1.1	Hose insertions: 5 hand	15
		3	0.717	2.15	3.6	locations (n = 15)	
		5	0.721	3.61	6.0		

average MVE torque of 8.07 Nm, and they present average effort durations of 1.12, 1.05, and 1.08 s, at $f = 15, 20$ and $25/\text{min}$, respectively (see their Table VII).

Snook, Ciriello, and Webster (1999) determined MATs for concentric wrist extension efforts with a pinch grip. Female respondents ($n = 20$) performed these efforts at $f = 15, 20$, and $25/\text{min}$. There were four training days, starting at 2 hr per day and ramping up to 6 hr per day. They present average effort durations of 0.64, 0.61, and 0.55 s, at $f = 15, 20$, and $25/\text{min}$, respectively (see their Table 4).

Ciriello, Webster, and Dempsey (2002) determined MATs for six tasks: (a) supination with a 31-mm screwdriver, (b) supination with a 40-mm screwdriver, (c) supination with a 39-mm yoke, (d) pronation with a 31-mm screwdriver, (e) ulnar deviation with a power grip, and (f) hand grip. Female respondents ($n = 10$) performed each task at $f = 15, 20$ and $25/\text{min}$. There were 5 days of training, ramping up to 7 hr per day. They collected MATs during 7-hr sessions from Days 6 thru to 23. I assumed the effort duration, for the 80 deg ulnar deviation task, to be the same as the average duration of 1.08 s (pooled across three frequencies) for an identical task in Snook et al. (1997; see their Table VII). The pronation and supination tasks were 90 deg, so I assumed they took 12.5% longer (1.215 s). The handgrip task was excluded, as sufficient information was not provided to allow for an estimate of effort durations.

Moore and Wells (2005) determined MATs for a wrist extension task using an in-line, powered screwdriver. Female respondents ($n = 8$) performed this task at $f = 3, 5, 10$, and $20/\text{min}$. This task was considered to be quasi-static as respondents were asked to isometrically resist the torque, but some small rotation did occur. They combined each frequency with three effort durations so that each had conditions with DCs of 0.25, 0.50, and 0.83, except $f = 20/\text{min}$, which did not have a DC = 0.25 condition. Thus, there were a total of 11 conditions tested in 15 sessions completed over 4 weeks. There were 3 days of training (1, 2, and 3 hr), and then they tested each of the 11 conditions on individual days, with one additional day for a biomechanical data collection. The testing days were 5 hr in

duration, with work for 50 min each hour and a 20-min break after the 3rd hour. I estimated MAT values from their Figure 4. They provided no maximum voluntary torque values in the article, so I estimated this to be 5.08 Nm, based on the combined data in Table 2 of Snook et al. (1995).

Potvin, Calder, Cort, Agnew, and Stephens (2006) determined MAFs for quasi-static pushing tasks (distance = 0.013 m) with the following three interfaces: (a) pulp pinch grip, (b) the finger tip, and (c) an oblique grasp, at $f = 2, 5$, and $7/\text{min}$. Female respondents ($n = 24$) performed the nine tasks in only one of three wrist postures: (a) neutral, (b) ulnar deviated, or (c) extended. There were 2 hr of training for each of the nine conditions, and then respondents determined MAFs and impulses over 1-hr sessions, on separate days, for each condition. There were no significant main or interaction effects of posture on MAF, so I averaged these values across postures, within each frequency/task combination. A video review after the study indicated that most respondents adopted a very different posture during the oblique grasp MVEs than they did during the prolonged oblique grasp psychophysical trials, such that appropriate strength values could not be determined for that task. The maximum voluntary forces for the pulp pinch push and finger tip push were 58.8 N and 56.6 N, respectively. The MAE values were 37.3, 36.3, and 33.3 Nm for pulp pinch push and 35.0, 32.9, 29.9 Nm for the finger tip push, at $f = 2, 5$, and $7/\text{min}$, respectively. Across all conditions, the average MAF and maximum acceptable impulse were 34.11 N and 5.45 Nm, respectively. I assumed rectangular force patterns and calculated duration by dividing the impulse by MAF and the duration of 0.160 was consistent across conditions.

Andrews et al. (2008) determined MAFs for isometric simulated hose insertion efforts with a power grip at $f = 1, 3$, and $5/\text{min}$. Female respondents ($n = 15$) performed isometric exertions for 0.75 s with five combinations of hand location and force direction: (a) long reach, medial force, (b) short reach, medial force, (c) push forward, (d) push down, and (e) pull back. Respondents received 15 hr of training, 1 hr for each combination of task and frequency, spread

over at least 5 days. They performed testing sessions of 1 hr, on different days, to determine MAF and impulse for each condition. The maximum voluntary forces were 87.6 N (medial, long), 113.6 N (medial, short), 124.3 N (push forward), 130.2 N (push down), and 163.6 N (pull back). Effort duration was estimated by dividing the average impulse by the average MAF, and when pooled across task, these were 0.685, 0.717, and 0.721 s for $f = 1, 3$, and 5/min, respectively.

Physiological Studies

All but 4 of the 69 psychophysical values included in this study had $DC \leq 0.50$. To supplement the psychophysical data for higher DC tasks, I also made comparisons with fatigue data from the five physiological studies outlined below. None of these studies exceeded 1 hr, and inclusion required that intermittent or sustained isometric contractions demonstrate physiological evidence of local muscle fatigue within 1 hr, as these tasks could definitely not be performed for 8 hr. Conditions not resulting in fatigue after 1 hr do not prove that the task could be performed for 8 hr, even with scheduled breaks.

Bjorksten and Jonsson (1977) studied intermittent and sustained isometric contractions of the elbow flexors with 4 males and 4 females. They found that the 1-hr endurance times for DCs of 0.30, 0.50, 0.70, and 1.00 had average effort magnitudes of 43.7%, 27.8%, 21.3%, and 7.9% of MVE, respectively. Sjogaard, Kiens, Jorgensen, and Saltin (1986) studied continuous 5% MVE isometric knee-extension task with 7 males and demonstrated muscle fatigue after 1 hr. Jorgensen, Fallentin, Krogh-Lund, and Jensen (1988) summarized five studies of sustained ($DC = 1.00$) isometric efforts of the finger flexors, elbow flexors, elbow extensors, and knee extensors, at 5%, 7%, and 10% MVE for up to 1 hr. They observed that only the 5% and 7% MVE contractions were held for 1 hr, but even these demonstrated subjective and objective evidence of localized muscle fatigue.

Bystrom and Kilbom (1990) studied intermittent and continuous isometric hand grip tasks, with 3 males and 3 females, at all 12 combinations of 10%, 25%, and 40% MVE with

DCs of 0.50, 0.67, 0.83, and 1.00. They observed evidence of physiological fatigue for 10% MVEs at $DC = 1.00$, 25% MVEs at $DC = 0.83$ and 1.00, and all 40% MVEs. Bystrom and Fransson-Hall (1994) also studied isometric handgrip tasks with 7 males and eight combinations of 10%, 25%, or 40% MVE, with DCs ranging from 0.25 to 1.00. They observed physiological evidence of fatigue for 10% MVEs at $DC = 1.00$, 25% MVEs at $DC = 0.67$, and 1.00 and 40% MVEs at $DC = 0.50, 0.67, 0.83$, and 1.00.

Analysis

There were a total of 69 MAF/MAT tasks, from the eight upper extremity studies, that met the inclusion criteria. For each value, I calculated the corresponding DC by multiplying the frequency by the effort duration and the average MAE by dividing the MAF or MAT by the corresponding maximum voluntary strength (MVE) value. A subsequent analysis was performed to determine if the MAE values had some systematic relationship with frequency, duration, and/or DC. A stepwise regression analysis was performed with $(1 - MAE)$ as the dependent variable and 21 independent variables which comprised (a) effort duration, (b) effort frequency, and (c) DC, each taken to the exponents 0.25, 0.50, 0.75, 1, 2, 3, and 4. The intercept was forced through zero.

The only variable needed in the original regression equation was $DC^{0.25}$. It was then assumed that the best exponent for DC was close to 0.25, and a second regression analysis was performed to determine the specific value between 0.05 and 0.45 in increments of 0.01. One 8-hr workday has 480 min, or 28,800 s. I assumed that it was acceptable to perform a single 1-s MVE per 8-hr workday. This averages to one effort per 0.0021 min and the DC for this is $DC = 1/28,800 = 0.0000347$ (0.00347%). Thus, I assumed that the MAE could be 100% at $DC \leq 0.0000347$ such that $(1 - MAE)$ was predicted with $(DC - 0.0000347)^E$. Exponents (E) were tested from 0.05 to 0.45, and the best was determined based on explained variance and RMS error for the 69 values included. A third regression was performed to determine the best equation with only

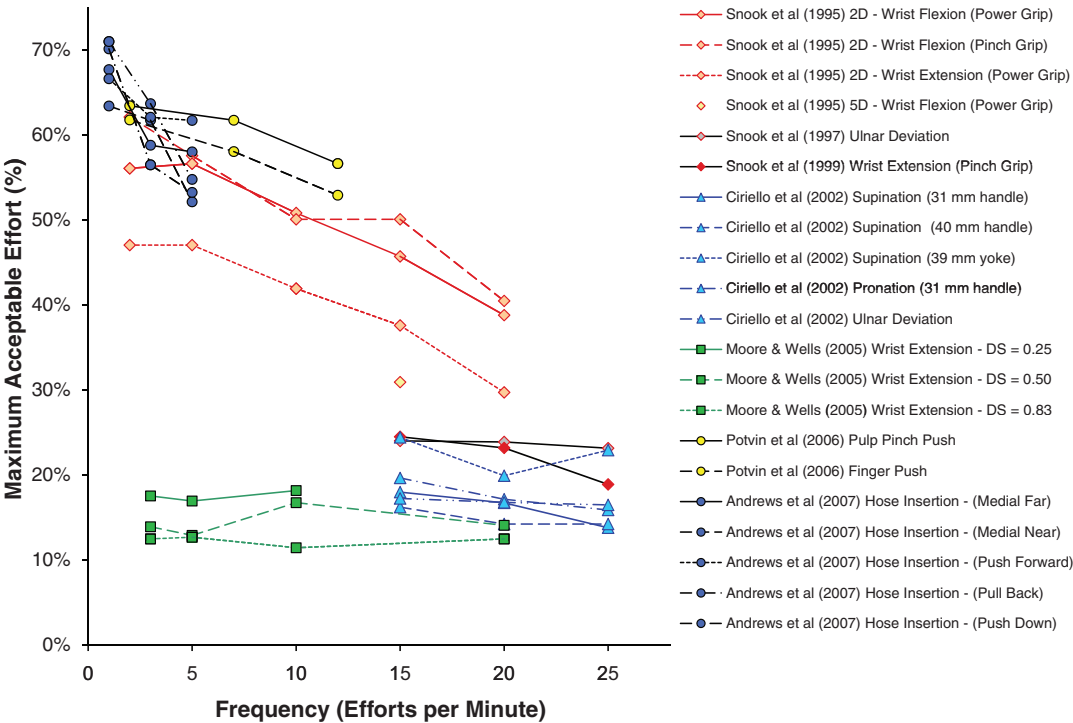


Figure 1. Maximum acceptable effort (as a percentage of maximum voluntary effort) versus frequency of effort. Mean data are shown for the upper extremity psychophysical studies meeting the inclusion criteria.

either frequency or duration as the independent variables.

RESULTS

Only a moderate relationship ($r^2 = .49$) existed between frequency and average MAE (Figure 1). In fact, at $f = 3/\text{min}$ the MAE values had a range of 51.2% MVE, and at $f = 15/\text{min}$ the range was 33.9% MVE. The best model, without DC as an input, included only frequency to the exponent 0.25 ($F = 1455, p < .001, r^2 = .47, \text{RMS error} = 14.6\% \text{ MVE}$).

There was a much stronger, nonlinear, negative relationship between DC and average MAE (Figure 2). The best exponent for DC was determined to be 0.24 ($F = 5404, p < .0001, r^2 = .87, \text{RMS error} = 7.2\% \text{ MVE}$) and the proposed equation takes the following form,

$$MAE = 1 - \left[DC - \frac{1}{28,800} \right]^{0.24}, \tag{1}$$

where

- MAE is average maximum acceptable effort, where 1.0 represents 100% MVE;
- DC is duty cycle, where 1.0 represents 100% of the cycle; and
- 28,880 represents the number of seconds in 8 hr. Its inverse is 0.0000347 and, when subtracted from DC, results in MAE = 1.0 (100%) for all $DC \leq 1 \text{ s/workday}$.

It should be noted that when the proposed equation is further simplified to Equation 2, the MAE values are within 0.6% MVE for all $DC > 0.0002$ (~6 s per 8 hr).

$$MAE = 1 - DC^{0.24}. \tag{2}$$

Given that little data exist to recommend MAEs at very high DCs, I decided to end the proposed curve at $DC = 0.90$ (Figure 3) and will address this further in the discussion. Although the predicted MAE is 100% of maximum at $DC = 0.0000347$ (1s per 8 hr), it decreases substantially to 66.9% MVE at a $DC = 0.01$ (0.6 s/min).

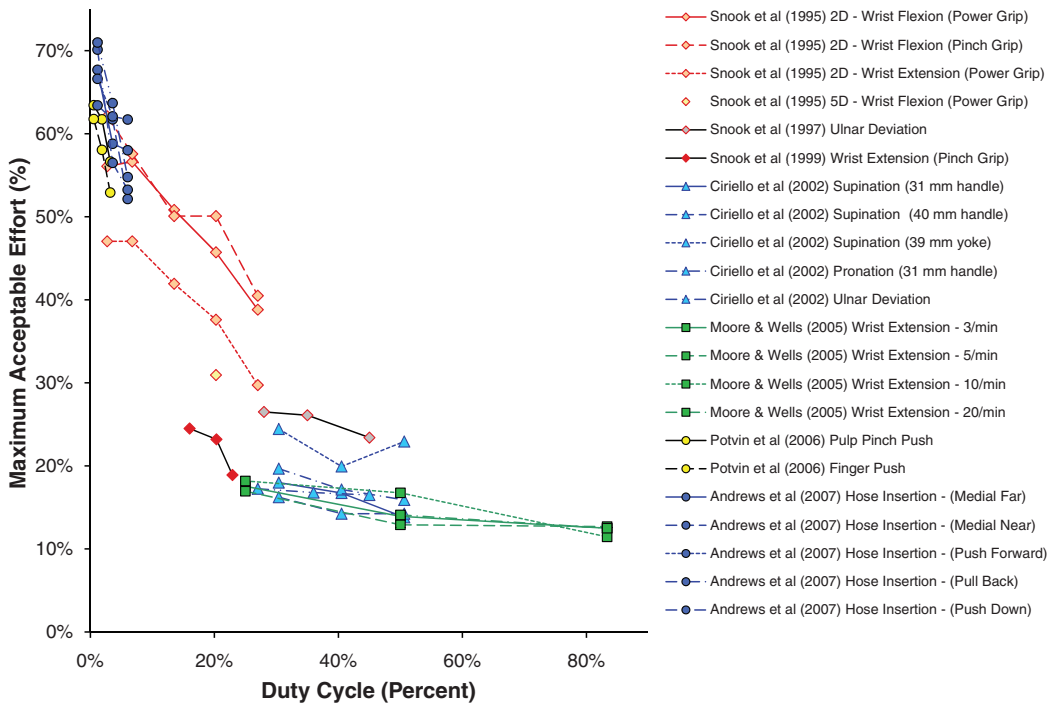


Figure 2. Maximum acceptable effort (as a percentage of maximum voluntary effort) versus duty cycle (percentage). Mean data are shown for the upper extremity psychophysical studies meeting the inclusion criteria.

The relationship between DC and MAE was quite robust, regardless of how the frequency and durations combined into DC. For example, Moore and Wells studied conditions of $f = 3, 5$, and $10/\text{min}$ with $DC = 0.25$, and the resulting MAEs were very similar at 17.5%, 17.0%, and 18.2% of maximum, respectively. Ciriello et al. (2002) had six conditions at $f = 15/\text{min}$ with $DC = 0.25$ and the mean MAE was 18.9%, whereas Snook et al. (1999) had one condition of $f = 25/\text{min}$ with $DC = 0.23$ and the MAE was also similar at 18.9%. Thus, for frequencies ranging from 3 to 26/min and effort durations ranging from 0.55 to 5 s, resulting in DCs of 0.23 to 0.25, the MAE remained consistently close to 18%.

Of the 69 tasks used, a total of 30 from Snook et al. (1995, $n = 15$) and Ciriello et al. (2002, $n = 15$) required a calculation to estimate effort duration based on similar tasks from that or previous studies. A sensitivity analysis was performed to determine the resulting impact if these estimates had been made to be 25% lower

or 25% higher. The results indicated very small changes in the relationship between DC and MAE, with the largest changes occurring with a 25% reduction in the estimated durations (r^2 decreased from .87 to .86 and RMS error increased from 7.2% to 8.1% MVE).

Once the MAE value is calculated, it can then be multiplied by the average MVE force or torque from the literature to determine the recommended average MAF or MAT (in N or Nm) for that DC (Equation 2).

$$MAF \text{ or } MAT = (MAE) \times (Maximum \text{ voluntary Strength in N or Nm}). \quad (3)$$

For example, Peebles and Norris (2003) measured an MVE strength of 70.0 ± 15.0 N for females (31–50 years) pulling a 20-mm block with a chuck pinch. If $DC = 0.25$, then the equation predicts the MAE to be 0.283, and the MAF, for the average female in that age range, is predicted to be $(0.283)(70.0) = 19.81$ N.

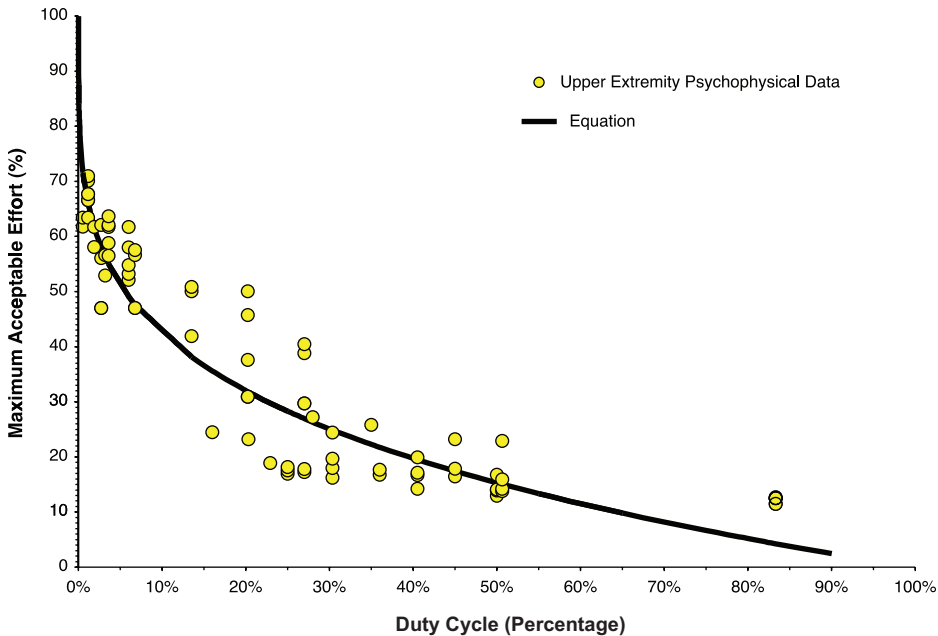


Figure 3. Maximum acceptable effort (as a percentage of maximum voluntary effort [MVE]) versus duty cycle as a percentage of an 8-hr workday. All upper extremity ($n = 69$) data are shown along with the curve from the proposed equation. Overall, the RMS difference was 7.2% MVE, and the $r^2 = .87$.

DISCUSSION

The main finding of this meta-analysis was that a strong, negative, exponential relationship exists between DC and MAE (normalized to MVE), based on previously published psychophysical data. This relationship was characterized with Equation 1 predicting MAE using DC as the only input variable. For all but extremely low DCs, the further simplified Equation 2 provides almost identical values. The relationship demonstrated a very rapid decline in MAE at low DCs, with a more gradual decline as DC increased to its higher values. The proposed Equation 1 performed very well for a variety of repetitive upper extremity force and torque tasks. Conversely, a much weaker relationship existed between frequency and MAE. Thus, the results demonstrate the importance of using DC, which combines duration and frequency (Figure 2), instead of just frequency (Figure 1), when predicting acceptable task demands in the workplace.

The primary advantage of the proposed equation is that it allows for the estimation of

the absolute magnitude of maximum acceptable repetitive forces and/or torques for tasks without specific published psychophysical and/or physiological data. In many cases, one can extract maximum strength values from the large database of published literature (e.g., DiDomenico & Nussbaum, 2003; Greig & Wells, 2004; Mital & Kumar, 1998; Peebles & Norris, 2003) or from available ergonomics software. However, in cases where published strength values are not available, the practitioner could even resort to collecting his or her own strength data from a group of workers on a simulation of the task being evaluated.

Equation 1 characterized MAEs through the full range of DCs from 0 to 0.90 and demonstrated strong predictive capabilities for a wide range of tasks demands, including wrist torques (flexion, extension and ulnar deviation with power and pinch grips), forearm torques (pronation and supination with screwdriver and yoke grips), and arm exertion forces (with pulp pinch, power grip, and finger tip push), with frequencies ranging from 1 to 25/min. The relationship

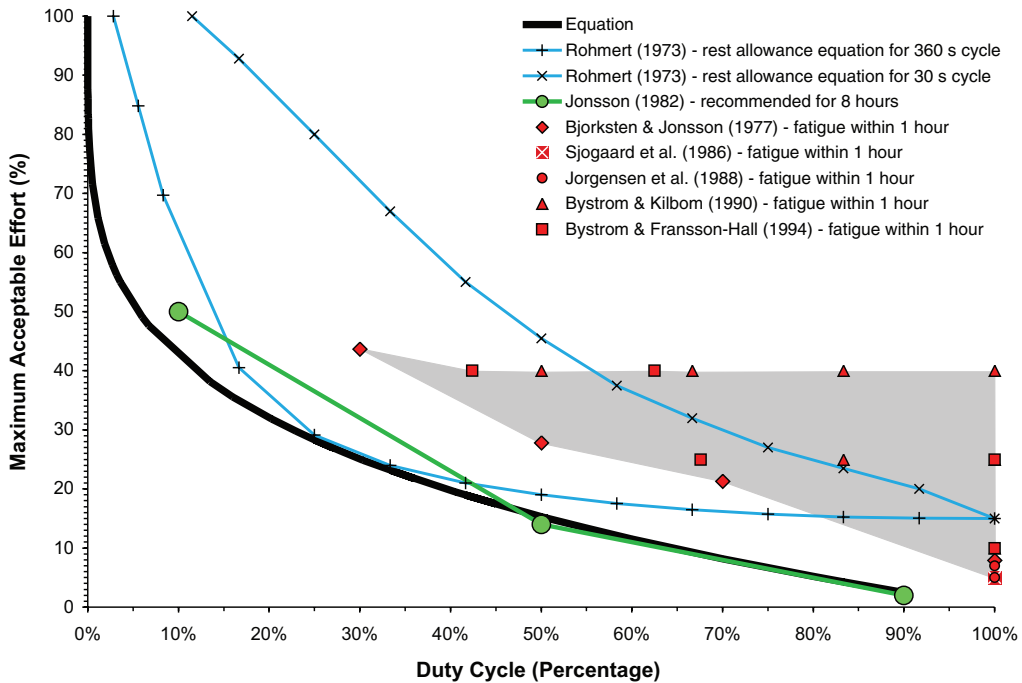


Figure 4. The proposed curve is plotted against some physiological data from the literature. The unlined symbols indicate conditions with physiological evidence of fatigue within 1 hr. The shaded area (and above) indicates a fatiguing region.

between DC and MVE was quite robust, regardless of how the frequency and duration combined into DC. This was somewhat unexpected, given that the need for recovery time increases exponentially with effort duration (Rohmert, 1973). However, the tasks studied in this meta-analysis had relatively short duration efforts, ranging from 0.16 to 16.67 s (Table 1). Thus, caution should be used when applying the equation to tasks with effort durations much greater than 17 s.

The proposed equation uses DCs calculated with the product of the average frequency of efforts and the measured (or estimated) average duration of each effort. This allows for a calculation of DC based only on task characteristics that can be observed and/or measured by an ergonomist. This does not estimate the actual DC of the individual muscles involved in the task, which may be active for longer than the task DC would suggest (Moore & Wells, 2005). However, I assumed that respondents in psychophysical studies would have been cognizant of this distinction and that the actual time course of muscle efforts was incorporated into their

experiences of discomfort and their decisions about acceptable demands. One limitation of DC, compared to frequency, is that accurate effort duration estimates can be difficult to collect and typically require direct measurement (Wells, Greig, & Ishac, 2007).

From a design perspective, the equation can inform the optimal combination of DC and MVE to maximize the net impulse of work, while still remaining within acceptable limits. Multiplying DC (% time) by the corresponding MAE (% of maximum force or torque), through the full range of DCs, creates a modified acceptable "impulse" curve, with an inverted U shape that has its maximum at DC = 0.40 (MAE = 19%). Thus, it is suggested that a DC of 0.40 may result in the maximum acceptable productivity.

A total of 65 of the 69 psychophysical data tasks used to develop the equation had DCs ≤ 0.50 . Thus, it was necessary to extrapolate the curve beyond DCs of 0.50. To provide support for this extrapolation, I compared the curve to published physiological data from higher DCs (Figure 4). Bjorksten and Jonsson (1977)

demonstrated physiological fatigue after 1 hr of intermittent contractions of the elbow flexors, with efforts of 27.8% of maximum at $DC = 0.50$ and 21.3% of maximum at $DC = 0.70$. Bystrom and Fransson-Hall (1994) also observed fatigue after 1 hr of hand gripping with efforts of 25% of maximum at $DC = 0.67$. For these same DCs of 0.50, 0.67, and 0.70, the proposed equation predicts acceptable demands of 15.3%, 9.2%, and 8.2% of maximum, respectively, for a full workday. Thus, based on these three DCs, the equation's MAEs were an average of 44% of the relative efforts demonstrating fatigue after 1 hr. Using the generalized endurance time equation proposed by Frey Law and Avin (2000), the theoretical relative efforts associated with endurance times of 8, 2, and 1 hr are 2.65%, 5.35%, and 7.60% of maximum, respectively, such that the ratio of 8:1 hr is 35%. If it is assumed that longest sustained contraction before a break would be 2 hr, the ratio of 2:1 hr is 50%. Given this comparison, and the fact that the physiological data do not necessarily represent the minimum effort that would cause fatigue within 1 hr, it appears that the corrections from the proposed equation are within the range that might be expected with theoretical endurance time curves. In addition, the slope of the proposed curve, from $DC = 0.30$ to 0.90, is in good agreement with that of the lower envelope of the region demonstrating physiological evidence of fatigue (Figure 4).

At DCs of 0.50 and 0.90, the proposed equation predicts values of $MAE = 15.3\%$ and 2.5% of maximum, respectively, which are in very good agreement with the corresponding values of 14% and 2% of maximum proposed by Jonsson (1982) for a full workday (Figure 4). However, for cycle times less than 1 min, the equation predicts values that are much lower than those calculated with the Rohmert (1973) rest allowance equations (Figure 4). The Rohmert equation would predict MAE values approaching those from the proposed equation only when cycle times are increased to 360 s, and this is the case only for DCs between 0.25 and 0.33 (Figure 4). For example, with a sustained contraction of 90 s, the Rohmert rest allowance is 270 s ($DC = 0.25$) for a 29.2% MVE effort, which is similar to the value from the proposed equation (Figure 4). However,

even with cycle times as high as 6 min, the Rohmert equation still estimates values that are much higher than those from both psychophysical studies and the proposed equation in the higher and lower ranges of DC.

The physiological data suggest that the demands for sustained efforts should be below 5% MVE (see Sjogaard et al., 1986, and Jorgensen et al., 1988 in Figure 4). This evidence disagrees with the 15% MVE that Rohmert (1973) assumed could be held indefinitely (Figure 4). Bjorksten and Jonsson (1977) suggest that this discrepancy is likely the result of the fact that Rohmert assumed that sustained efforts, which could be maintained for 15 min, could then be maintained indefinitely. They further suggest that the acceptable MVE value, for sustained efforts of 4 to 8 hr, might be as low as a small percentage of maximum and state that completely sustained efforts "should never be accepted in a work process no matter how weak the contraction" (p. 27). Veiersted and Westgaard (1993) postulated that long sustained efforts, even at low levels, might be an important risk factor for muscle pain and injury. They found that symptoms were more prevalent with tasks having sustained efforts at 1.9% versus 1.3% MVE (from Kilbom, 1994). Furthermore, Veiersted, Westgaard, and Andersen (1990) and Jensen, Nilsen, Hansen, and Westgaard (1993) found that sustained levels as low as 0.5% to 1% MVE might be associated with shoulder and/or neck pain. Hagg (1991) proposed that low-level, static contractions result in the tonic recruitment of the same low threshold Type I motor units ("Cinderella units") such that those fibers can become overloaded over prolonged durations. Veiersted and Westgaard (1993) and Hagg (2000) propose a number of physiological abnormalities that can result from this type of loading, including sensitization of nociceptors and mitochondrial disturbances. Sjogaard and Jensen (1999) suggest that the goal in workplace design should be the optimization of mechanical workload, instead of its minimization. However, they also suggest that work cycles should have at least some periods of complete relaxation. Sjogaard, Lundberg, and Kadefors (2000) state, "[I]t is possible that the lack of relaxation is an even more important health risk than is

the absolute level of muscle contraction or the frequency of muscle activation" (p. 104). Conversely, Aaras (1987) concluded that a 1% MVE continuous effort of the trapezius muscle was acceptable "for most of the work day," if adequate breaks were allowed.

It is apparent that definitive evidence is not available in the literature to confidently recommend an acceptable level of effort for prolonged, sustained efforts. As such, I decided to end the proposed curve at $DC = 0.90$, where the MAE is 2.5%. One might assume that acceptable efforts at $DC > 0.90$ would not exceed those at $DC = 0.90$. However, given the lack of consensus in the literature, it is left to the readers to decide for themselves how the curve should progress at $DC > 0.90$. Having said this, it is highly unlikely that an ergonomist will ever come across a repetitive task where the product of frequency and effort duration even comes close to $DC \geq 0.90$. There may be cases where workers maintain sustained contractions without rest, but such work conditions are outside the scope of the equation being proposed for repetitive tasks.

It is important that those intending to use this equation understand its limitations and constraints. The equation is intended for use with relatively stereotypical tasks having similar characteristics. There will always be some variability in external demand, in part because of quality issues (size, stiffness, etc.) and changes in worker performance, but I suggest that the average force or torque can be used in cases when the demand does not vary substantially. The equation is limited in its application to individual tasks and cannot be extrapolated for more complex work conditions with multiple task elements. The user is cautioned that multiple ergonomic assessments, demonstrating that individual elements are acceptable in isolation, do not necessarily ensure that the entire combined task is acceptable for the target population. Finally, in developing the equation, I assumed that MVEs are acceptable only when performed very infrequently because such efforts are, by definition, voluntary and would typically involve some protective inhibition. However, there are limited data, at the very low DCs, to completely validate the exact shape of the curve in that small region.

This equation was developed to assist ergonomic practitioners to estimate acceptable forces and torques in the workplace and not to theoretically model the underlying mechanisms governing these limits. Ergonomists are often required to recommend acceptable task loads but have limited methods to do so, especially for the types of repetitive tasks that are very prevalent in many occupational environments. I attempted to fit an equation to empirical psychophysical data, in much the same way that the widely used National Institute for Occupational Safety and Health lifting equation (Waters et al., 1993) was developed, in part, to fit the data of a variety of lifting psychophysical studies (e.g., Ayoub et al., 1978; Snook & Ciriello, 1991).

Although Equation 1 had an explained variance of 87% and an RMS error of 7.2% MVE, there were over- and underpredictions of average MAEs as high as 11.4% and 18.2% MVE, respectively. Thus, it appears that an equation based only on DC, although simple and convenient to use, does not completely capture all of the variables that influence the MAE. Also, given the scarcity of published upper extremity psychophysical studies using male respondents, only female data contributed to the development of current equation. Thus, it should be used only for determining female MAE values. Across the eight studies used, there were a total of 117 respondents represented. However, three of the studies had relatively small respondent numbers of 8, 10, and 11 (Table 1), and it is possible that larger respondent groups may have affected their average MAEs and the resulting equation. Clearly, more research is needed to (a) further validate the equation with a wider variety of upper extremity tasks, (b) validate the equation for use with male respondents or determine a unique relationship between DC and MAE for males, (c) determine MAE values at DCs above 0.50, and (d) determine the additional variables that would further improve the prediction of MAE for any combination of task parameters.

CONCLUSION

Based on a meta-analysis of existing upper extremity psychophysical data, a strong, negative, exponential relationship exists between MAE, as a percentage of maximum, and DC (but

not frequency). The proposed equation predicts MAE using DC as the only input variable. It performed very well for a wide variety of repetitive upper extremity task demands. The equation outputs can be multiplied by maximum strength data (from the literature or directly measured) to predict the absolute magnitude of MAFs or MATs, for repetitive tasks with DCs up to 0.90.

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KEY POINTS

- For existing upper extremity psychophysical data, duty cycle has a strong relationship with maximum acceptable effort, normalized to maximum.
- The proposed equation accurately characterizes this relationship.
- At higher duty cycles, the equation also demonstrated good agreement with existing physiological data.
- This equation predicts maximum acceptable efforts, for a wide variety of repetitive occupational tasks, in cases where psychophysical or physiological data do not currently exist.
- The absolute magnitude of maximum acceptable forces and torques can be predicted using the equation outputs to scale existing maximum strength data.

REFERENCES

- Aaras, A. (1987). Postural load and the development of musculoskeletal illness. *Scandinavian Journal of Rehabilitation Medicine*, 18(suppl.), 5–35.
- Andrews, D., Potvin, J., Calder, C. I., Cort, J., Agnew, M., & Stephens, A. (2008). Acceptable peak forces and impulses during manual hose insertions in the automobile industry. *International Journal of Industrial Ergonomics*, 38, 193–201.
- Ayoub, M. M., Bethea, N. J., Deivanayagam, S., Asfour, S. S., Bakken, G. M., & Liles, D. (1978). *Determination and modelling of lifting capacity* (Final report, Grant No. 5-R01-OH-00545-02). Washington, DC: U.S. Department of Health and Human Services, National Institute for Occupational Safety and Health.
- Ayoub, M. M., & Mital, A. (1989). *Manual materials handling*. London, UK: Taylor & Francis.
- Bjorksten, M., & Jonsson, B. (1977). Endurance limit of force in long-term intermittent static contractions. *Scandinavian Journal of Work, Environment & Health*, 3, 23–27.
- Bystrom, S., & Fransson-Hall, C. (1994). Acceptability of intermittent handgrip contractions based on physiological response. *Human Factors*, 36, 158–171.
- Bystrom, S., & Kilbom, A. (1990). Physiological response in the forearm during and after isometric intermittent handgrip. *European Journal of Applied Physiology and Occupational Physiology*, 60, 457–466.
- Ciriello, V., Snook, S., Webster, B., & Dempsey, P. (2001). Psychophysical study of six hand movements. *Ergonomics*, 44, 922–936.
- Ciriello, V. M., Webster, B. S., & Dempsey, P. G. (2002). Maximal acceptable torques of highly repetitive screw driving, unlar deviation, and handgrip tasks for 7-hour workdays. *American Industrial Hygiene Association Journal*, 63, 594–604.
- DiDomenico, A., & Nussbaum, M. (2003). Measurement and prediction of single and multi-digit finger strength. *Ergonomics*, 46, 1531–1548.
- Frey Law, L. A., & Avin, K. G. (2000). Endurance time is joint-specific: A modelling and meta-analysis investigation. *Ergonomics*, 53, 109–129.
- Greig, M., & Wells, R. (2004). Measurement of prehensile grasp capabilities by a force and moment wrench: Methodological development and assessment of manual workers. *Ergonomics*, 47, 41–58.
- Hagg, G. M. (1991). Static work loads and occupational myalgia—A new explanation model. In P. A. Anderson, D. J. Hobart, & J. V. Danof (Eds.), *Electromyographical kinesiology* (pp. 141–144). Amsterdam, Netherlands: Elsevier.
- Hagg, G. (2000). Human muscle fibre abnormalities related to occupational load. *European Journal of Applied Physiology*, 83, 159–165.
- Jensen, C., Nilsen, K., Hansen, K., & Westgaard, R. H. (1993). Trapezius muscle load as a risk indicator for occupational shoulder-neck complaints. *International Archives of Occupational and Environmental Health*, 64, 415–423.
- Jonsson, B. (1982). Measurement and evaluation of local muscular strain in the shoulder during constrained work. *Journal of Human Ergology*, 11, 73–88.
- Jorgensen, K., Fallentin, N., Krogh-Lund, C., & Jensen, B. (1988). Electromyography and fatigue during prolonged, low-level static contractions. *European Journal of Applied Physiology*, 57, 316–321.
- Kilbom, A. (1994). Repetitive work of the upper extremity: Part 2—The scientific basis (knowledge base) for the guide. *International Journal of Industrial Ergonomics*, 14, 151–178.
- Kim, C. H., & Fernandez, J. E. (1993). Psychophysical frequency for a drilling task. *International Journal of Industrial Ergonomics*, 12, 209–218.
- Mital, A., & Kumar, S. (1998). Human muscle strength definitions, measurement, and usage: Part 1—Guidelines for the practitioner. *Industrial Ergonomics*, 22, 101–121.
- Moore, A., & Wells, R. (2005). Effect of cycle time and duty cycle on psychophysically determined acceptable levels in a highly repetitive task. *Ergonomics*, 48, 859–873.
- Muppasani, A. K. L., & Fernandez, J. E. (1996). The effect of adjustment period on maximum acceptable frequency for a drilling task. In J. G. Chen & A. Mital (Eds.), *Advances in industrial engineering applications and practice* (pp. 1133–1138). Vienna, VA: International Journal of Industrial Engineering.
- Peebles, L., & Norris, B. (2003). Filling in “gaps” in strength data for design. *Applied Ergonomics*, 34, 73–88.
- Potvin, J. R., Calder, I. C., Cort, J., Agnew, M. J., & Stephens, A. (2006). Maximal acceptable forces for manual insertions using a pulp pinch, oblique grasp and finger press. *International Journal of Industrial Ergonomics*, 36, 779–787.

- Potvin, J. R., Chiang, J., Mckean, C., & Stephens, A. (2000). A psychophysical study to determine acceptable limits for repetitive hand impact severity during automotive trim installation. *International Journal of Industrial Ergonomics*, 26, 625–637.
- Rohmert, W. (1973). Problems in determining rest allowances. Part I: Use of modern methods to evaluate stress and strain in static muscular work. *Applied Ergonomics*, 4, 91–95.
- Sjogaard, G., & Jensen, B. R. (1999). Low-level static exertions. In W. Karwowski & W. S. Marras (Eds.), *The occupational ergonomics handbook* (pp. 247–259). London, UK: CRC Press.
- Sjogaard, G., Kiens, B., Jorgensen, K., & Saltin, B. (1986). Intramuscular pressure, EMG and blood flow during low-level prolonged static contraction in man. *Acta Physiologica Scandinavica*, 128, 475–484.
- Sjogaard, G., Lundberg, U., & Kadefors, R. (2000). The role of muscle activity and mental load in the development of pain and degenerative processes at the muscle cell level during computer work. *European Journal of Applied Physiology*, 83, 99–105.
- Snook, S. (1968). Maximum frequency of lift acceptable to male industrial workers. *American Industrial Hygiene Association Journal*, 29, 531–536.
- Snook, S. H. (1999). Future directions of psychophysical studies. *Scandinavian Journal of Work, Environment & Health*, 25(Suppl. 4), 13–18.
- Snook, S. H., & Ciriello, V. M. (1991). The design of manual handling tasks: Revised tables of maximum acceptable weights and forces. *Ergonomics*, 34, 1197–1213.
- Snook, S. H., Ciriello, V. M., & Webster, B. S. (1999). Maximum acceptable forces for repetitive wrist extension with a pinch grip. *International Journal of Industrial Ergonomics*, 24, 579–590.
- Snook, S. H., Vaillancourt, D. R., Ciriello, V. M., & Webster, B. S. (1995). Psychophysical studies of repetitive wrist flexion and extension. *Ergonomics*, 38, 1488–1507.
- Snook, S. H., Vaillancourt, D. R., Ciriello, V. M., & Webster, B. S. (1997). Maximum acceptable forces for repetitive ulnar deviation of the wrist. *American Industrial Hygiene Association Journal*, 58, 509–517.
- Stevens, S. S. (1957). On the psychophysical law. *Psychological Review*, 64, 153–181.
- Veiersted, K. B., & Westgaard, R. H. (1993). Development of trapezius myalgia among female workers performing light manual work. *Scandinavian Journal of Work, Environment & Health*, 19, 277–283.
- Veiersted, K., Westgaard, R., & Andersen, P. (1990). Pattern of muscle activity during stereotyped work and its relation to muscle pain. *International Archives of Occupational and Environmental Health*, 62, 31–41.
- Waters, T. R., Putz-Anderson, V., Garg, A., & Fine, L. J. (1993). Revised NIOSH equation for the design and evaluation of manual lifting tasks. *Ergonomics*, 36, 749–776.
- Wells, R., Greig, M., & Ishac, M. (2007). *Predicting distal arm demand from task requirements*. (SAE Technical Paper 2007-01-2509).
- Wu, S., & Chen, J. (2003). Effects of the adjustment period on psychophysically determined maximum acceptable weight of lift and the physiological cost. *International Journal of Industrial Ergonomics*, 31, 287–294.

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Corrigendum

Potvin, J. R. (2012). Predicting maximum acceptable efforts for repetitive tasks: An equation based on duty cycle. *Human Factors*, 54, 175–187. Original DOI: 10.1177/0018720811424269

In the prepublication (OnlineFirst) version of “Predicting Maximum Acceptable Efforts for Repetitive Tasks: An Equation Based on Duty Cycle” by Jim R. Potvin, the references to *pronation* and *supination* from Ciriello, Webster, and Dempsey (2002) were erroneously reversed in Table 1, Figures 1 and 2, and page 5. There are three supination conditions and one pronation condition.

The errors have been corrected in the online and print versions of this paper, which will appear in the April 2012 issue, Volume 54, Number 2.